Sanding Machine



Category: **Power Tools & Workshop Machines** Type: **Full SOP** Priority: **High** 2 labs

Covers these lab items: Sanders/Grinders; Sanding Machine

What it is used for

A sanding machine smooths surfaces, removes coatings, or prepares materials using an abrasive belt, disc, pad, or wheel. Select the correct abrasive and operating speed for the workpiece and control dust and contact hazards throughout the task.

Main parts

- Motor and machine housing
- Sanding belt, disc, pad, or abrasive wheel
- Work support or backing plate
- Power switch and speed control, if fitted
- Dust extraction port or collection bag
- Protective guard and handle

Before you start

1. Identify the sanding machine type and read its operating instructions.
2. Inspect the abrasive, backing pad, guards, handle, cord, and plug for damage.
3. Confirm the abrasive grade and speed rating suit the workpiece and tool.
4. Secure the workpiece using a clamp or suitable fixture.
5. Connect approved dust extraction and ensure the work area is ventilated.
6. Wear eye protection, hearing protection, and suitable respiratory protection for the dust.

How to use

1. Disconnect power before installing or changing the abrasive attachment.
2. Fit the correct abrasive securely and align it according to the instructions.
3. Check that guards and dust extraction components are fitted correctly.
4. Position the machine firmly and keep the abrasive clear of the workpiece before starting.
5. Connect the tool to a suitable power supply and switch it on.
6. Allow the machine to reach its normal operating speed before contacting the workpiece.
7. Apply steady, controlled pressure and keep the machine moving as required.
8. Avoid excessive pressure that may overheat the motor or damage the surface.
9. Monitor the finish, dust extraction, vibration, and tool temperature during operation.
10. Switch off and wait for all moving parts to stop before setting the tool down.
11. Disconnect power before clearing a blockage, changing abrasive, or performing maintenance.
12. Remove dust safely and inspect the finished surface and machine.

⚠ Safety

- Only trained users should operate the sanding machine.
- Never touch the moving abrasive or remove guards during operation.
- Do not use damaged, cracked, incorrectly fitted, or over-speed-rated abrasives.
- Secure loose workpieces and keep hands away from the sanding area.
- Control dust, especially from wood, coatings, metals, or other hazardous materials.
- Disconnect power before adjustment, cleaning, or changing accessories.

Care & storage

- Remove dust from vents and the machine exterior after use.
- Inspect the abrasive, backing pad, belt tracking, and guards regularly.
- Replace worn abrasives and damaged components with compatible approved parts.
- Keep electrical connections dry and store the machine in a clean location.
- Arrange servicing if the tool overheats, vibrates excessively, or produces unusual noise.

Common mistakes to avoid

- Using an abrasive with the wrong grade or speed rating.
- Applying excessive pressure and overheating the workpiece.
- Operating without suitable dust extraction or respiratory protection.
- Changing abrasives or clearing jams while the machine is connected to power.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

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